



Dementia with Lewy Bodies and Parkinson Disease Dementia

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Lewy body dementia includes clinically diagnosed dementia with Lewy bodies and Parkinson disease dementia. **Dementia with Lewy bodies** is chronic cognitive deterioration characterized by cellular inclusions called Lewy bodies in the cytoplasm of cortical neurons. **Parkinson disease dementia** is cognitive deterioration characterized by Lewy bodies in the substantia nigra; it develops late in Parkinson disease.

(See also [Overview of Delirium and Dementia](#) and [Dementia](#).)

Dementia is chronic, global, usually irreversible deterioration of cognition.

Dementia with Lewy bodies is the third most common dementia. Age of onset is typically > 60 years.

Lewy bodies are spherical, eosinophilic, neuronal cytoplasmic inclusions composed of aggregates of alpha-synuclein, a synaptic protein. They occur in the cortex of patients who have dementia with Lewy bodies. Neurotransmitter levels and neuronal pathways between the striatum and the neocortex are abnormal.

Lewy bodies also occur in the substantia nigra of patients with [Parkinson disease](#), and dementia (Parkinson disease dementia) may develop late in the disease. Parkinson disease dementia has been reported in about 25 to 30% of patients with Parkinson disease, rising to a prevalence of 83% in people who have had Parkinson disease for at least 20 years ([1](#)).

Because Lewy bodies occur in dementia with Lewy bodies and in Parkinson disease dementia, some experts think the 2 disorders may be part of a more generalized synucleinopathy affecting the central and peripheral nervous systems. Lewy bodies sometimes occur in patients with [Alzheimer disease](#), and patients with dementia with Lewy bodies may have neuritic plaques and neurofibrillary tangles. Dementia with Lewy bodies, Parkinson disease, and Alzheimer disease overlap considerably. Further research is needed to clarify the relationships among them.

Both dementia with Lewy bodies and Parkinson disease dementia have a progressive course with a poor prognosis.

[Dementia](#) should not be confused with [delirium](#), although cognition is disordered in both. The following usually helps distinguish dementia from delirium:

- **Dementia** affects mainly memory, is typically caused by anatomic changes in the brain, has slower onset, and is generally irreversible.
- **Delirium** affects mainly attention, is typically caused by acute illness or drug toxicity (sometimes life threatening), and is often reversible.

Other specific characteristics also help distinguish the dementia and delirium (see table [Differences Between Delirium and Dementia](#)).

Reference

1. [Aarsland D, Creese B, Politis M, et al.](#) Cognitive decline in Parkinson disease. Nat Rev Neurol. 2017;13(4):217-231. doi:10.1038/nrneurol.2017.27

Symptoms and Signs

Dementia with Lewy bodies

Initial cognitive deterioration in dementia with Lewy bodies resembles [that in other dementias](#). However, dementia with Lewy bodies often manifests with early and prominent deficits in attention, executive function, and visuospatial ability; prominent or persistent memory impairment tends to occur as the dementia progresses.

Extrapyramidal symptoms (typically including rigidity, bradykinesia, and gait instability) occur. However, in dementia with Lewy bodies (unlike in Parkinson disease), cognitive and extrapyramidal symptoms usually begin within 1 year of each other. Also, the extrapyramidal symptoms differ from those of Parkinson disease; in dementia with Lewy bodies, tremor does not occur early, rigidity of axial muscles with gait instability occurs early, and deficits tend to be symmetric. Repeated falls are common.

Fluctuating cognitive function is a relatively specific feature of dementia with Lewy bodies. Periods of being alert, coherent, and oriented may alternate with periods of being confused and unresponsive to questions, usually over a period of days to weeks but sometimes during the same interview.

Memory is impaired, but the impairment appears to result more from deficits in alertness and attention than in memory acquisition; thus, short-term recall is affected less than digit span memory (ability to repeat 7 digits forward and 5 backward).

Patients may stare into space for long periods. Excessive daytime drowsiness is common.

Visuospatial and visuoconstructional abilities (tested by block design, clock drawing, or figure copying) are affected more than other cognitive deficits.

Visual **hallucinations** are common and often threatening, unlike the benign hallucinations of Parkinson disease. Auditory, olfactory, and tactile hallucinations are less common. Delusions occur in 50 to 65% of patients and are often complex and bizarre, compared with the simple persecutory ideation common in Alzheimer disease.

Autonomic dysfunction is common, and unexplained syncope may result. Autonomic dysfunction may occur simultaneously with, or after, onset of cognitive deficits. Extreme sensitivity to antipsychotics is typical.

Sleep problems are common. Many patients have [rapid eye movement \(REM\) sleep behavior disorder](#), a parasomnia characterized by vivid dreams without the usual physiologic paralysis of skeletal muscles during REM sleep. As a result, dreams may be acted out, sometimes injuring the bed partner.

Parkinson disease dementia

In Parkinson disease dementia (unlike in dementia with Lewy bodies), cognitive impairment that leads to dementia typically begins 10 to 15 years after the onset of motor symptoms.

Parkinson disease dementia may affect multiple cognitive domains including attention, memory, and visuospatial, constructional, and executive functions. Executive dysfunction typically occurs earlier and is more common in Parkinson disease dementia than in Alzheimer disease.

Psychiatric symptoms (eg, hallucinations, delusions) appear to be less frequent and/or less severe than in dementia with Lewy bodies.

In Parkinson disease dementia, postural instability and gait abnormalities are more common, motor decline is more rapid, and falls are more frequent than in Parkinson disease without dementia.

Diagnosis

- Clinical criteria
- Neuroimaging to exclude other disorders

Diagnosis is clinical, but sensitivity and specificity are poor.

A general [diagnosis of dementia](#) requires all of the following:

- Cognitive or behavioral (neuropsychiatric) symptoms interfere with the ability to function at work or do usual daily activities.
- These symptoms represent a decline from previous levels of functioning.
- These symptoms are not explained by delirium or a major psychiatric disorder.

Evaluation of cognitive function involves taking a history from the patient and from someone who knows the patient plus doing a bedside [mental status examination](#) or, if bedside testing is inconclusive, [formal neuropsychologic testing](#).

Diagnosis of dementia with Lewy bodies is considered probable if 2 of the following 4 core features are present and is considered possible if only one is present:

- Fluctuations in cognition
- Visual hallucinations
- Rapid eye movement (REM) sleep behavior disorder
- Parkinsonism

Supportive evidence includes repeated falls, syncope, autonomic dysfunction, excessive daytime sleepiness, and sensitivity to antipsychotics.

Overlap of symptoms in dementia with Lewy bodies and Parkinson disease dementia may complicate diagnosis:

- When motor deficits (eg, tremor, bradykinesia, rigidity) precede and are more severe than cognitive impairment, Parkinson disease dementia is usually diagnosed.
- When early cognitive impairment (particularly executive dysfunction) and behavioral disturbances predominate, dementia with Lewy bodies is usually diagnosed.

Patients with dementia with Lewy bodies often have impaired alertness, which is more characteristic of delirium than dementia; thus, [evaluation for delirium](#) should be done, particularly for common causes such as

- Drugs and medications, particularly anticholinergics, psychoactive drugs, and opioids
- Dehydration
- Infection

Other causes of dementia should also be excluded.

The parkinsonian features in dementia with Lewy bodies (eg, axial rigidity, falls) may occur in early stages of other neurodegenerative disorders (eg, [progressive supranuclear palsy](#) [PSP]). People with PSP may also develop subcortical dementia. Identifying other characteristic features of PSP, such as impairment of vertical eye movements and [midbrain atrophy on midsagittal MRI](#) (hummingbird or King Penguin sign), may help differentiate PSP from dementia with Lewy bodies.

CT and MRI show no characteristic changes in Lewy body dementia but can help exclude other causes of dementia. Single-photon emission CT (SPECT) of the brain with 123I-FP-CIT (*N*-3-fluoropropyl-2-beta-carbomethoxy-3beta-[4-iodophenyl]-tropane), a radioactive tracer of dopamine transporter, can help detect basal ganglia dopaminergic neuronal dysfunction and function as a biomarker for diagnosing dementia with Lewy bodies.

Other imaging tests that may help identify dementia with Lewy bodies include

- Positron emission tomography (PET) of the brain with fluorine-18 (18F)-labeled deoxyglucose (fluorodeoxyglucose, or FDG)
- Myocardial scintigraphy with iodine 123-meta-iodobenzylguanidine (MIBG); in dementia with Lewy bodies, cardiac uptake of MIBG is reduced because the postganglionic cardiac

sympathetic system is denervated

However, these tests are not routinely used in clinical practice.

Confirmation of REM sleep behavior disorder using polysomnography may help establish the diagnosis of dementia with Lewy bodies.

Definitive diagnosis requires autopsy samples of brain tissue.

Treatment

- Supportive care
- Sometimes medications

Treatment for dementia with Lewy bodies and Parkinson disease dementia is generally [supportive](#). For example, the environment should be bright, cheerful, and familiar, and it should be designed to reinforce orientation (eg, placement of large clocks and calendars in the room). Measures to ensure patient safety (eg, signal monitoring systems for patients who wander) should be implemented.

Troublesome cognitive and behavioral symptoms, such as hallucinations, paranoid thoughts, irritability, and aggressiveness, are treated with medications only when other measures are ineffective.

Medications

[Cholinesterase inhibitors](#) may improve cognitive function and be helpful in patients who have dementia with Lewy bodies or Parkinson disease dementia. [Rivastigmine](#), or other cholinesterase inhibitors, may be initiated at a low oral dose and gradually titrated upward as needed.

In about half of patients with dementia with Lewy bodies, extrapyramidal symptoms respond to [antiparkinsonian medications](#), but psychiatric symptoms may worsen. If such medications are needed, [levodopa](#) is preferred.

In dementia with Lewy bodies, traditional antipsychotics, even at very low doses, tend to acutely worsen extrapyramidal symptoms and are best avoided.

[Pimavanserin](#), a nondopaminergic selective inverse agonist of the serotonin 5-HT-2A receptor, can be used to treat psychosis (hallucinations, delusions) in patients with Parkinson disease dementia.

End-of-life issues

Because insight and judgment deteriorate in patients with dementia, appointment of a family member, guardian, or lawyer to oversee finances may be necessary. Early in dementia, before the patient is incapacitated, the patient's wishes about care should be clarified, and financial and legal arrangements (eg, durable power of attorney, [durable power of attorney for health care](#)) should be made. When these documents are signed, the patient's [capacity](#) should be evaluated, and evaluation results recorded. Decisions about artificial feeding and treatment of acute disorders are best made before the need develops.

In advanced dementia, [palliative measures](#) may be more appropriate than highly aggressive interventions or hospital care.

Key Points

- Because Lewy bodies occur in dementia with Lewy bodies and in Parkinson disease, some experts hypothesize that the 2 disorders are part of the same synucleinopathy affecting the central and peripheral nervous systems.
- Suspect dementia with Lewy bodies if dementia develops nearly simultaneously with parkinsonian features and when dementia is accompanied by fluctuations in cognition, loss of attention, vivid dreams, psychiatric symptoms (eg, visual hallucinations; complex, bizarre delusions), and autonomic dysfunction.
- Suspect Parkinson disease dementia if dementia begins years after parkinsonian features, particularly if executive dysfunction occurs early.
- Consider use of [rivastigmine](#) and sometimes other cholinesterase inhibitors to try to improve cognition in patients who have dementia with Lewy bodies or Parkinson disease dementia.

Drug Information for the Topic



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